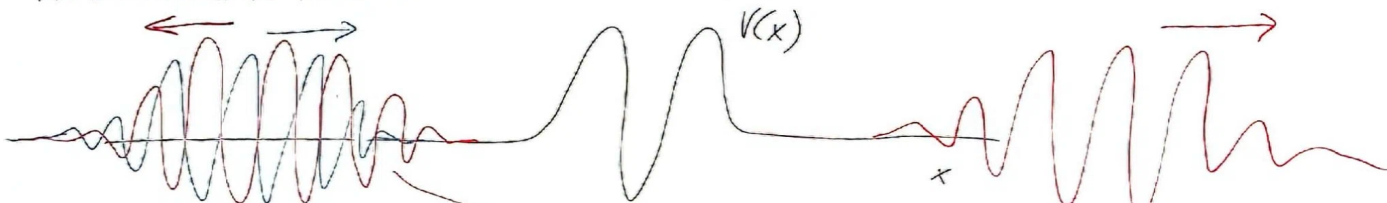


TRANSMISSION AND REFLECTION PHENOMENA



$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{in}\rangle \quad (t \rightarrow -\infty)$$

$$|\phi_{in}\rangle = \int_{-\infty}^{+\infty} dp \phi_{in}(p) |\psi_p^F\rangle$$

$$\phi_{in}(p) = \langle \psi_p^F | \phi_{in} \rangle$$

$$|\phi_{out}\rangle = \hat{S} |\phi_{in}\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{out}\rangle \quad (t \rightarrow +\infty)$$

$$|\phi_{out}\rangle = \int_{-\infty}^{+\infty} dp \phi_{out}(p) |\psi_p^F\rangle$$

$$\phi_{out}(p) = \langle \psi_p^F | \phi_{out} \rangle$$

$$\langle \psi_p^F | \hat{S} | \psi_{p'}^F \rangle = \delta(p-p') - 2\pi i \delta\left(\frac{p^2}{2m} - \frac{p'^2}{2m}\right) \langle \psi_p^F | \hat{T}(E) | \psi_{p'}^F \rangle$$

$$\hat{T}(E) = \hat{V} + \hat{V} \frac{1}{E - \hat{H} + i\epsilon} \hat{V}$$

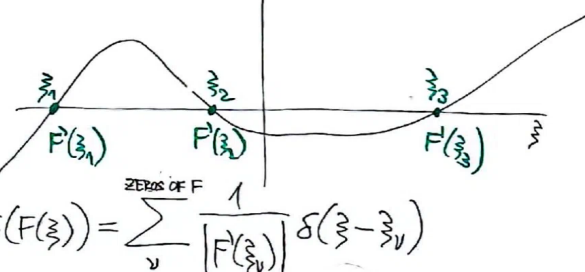
$$E = \frac{p^2}{2m} = \frac{p'^2}{2m}$$

$$\rightarrow \langle \psi_p^F | \hat{V} | \psi_{p'}^F \rangle$$

$$\begin{aligned} |\phi_{out}\rangle &= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle \langle \psi_p^F | \hat{S} | \psi_{p'}^F \rangle \langle \psi_{p'}^F | \phi_{in} \rangle = \\ &= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle \delta(p-p') \phi_{in}(p) + \\ &+ \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle 2\pi i \delta\left(\frac{p^2}{2m} - \frac{p'^2}{2m}\right) \langle \psi_p^F | \hat{V} | \psi_{p'}^F \rangle \phi_{in}(p') = \\ &= \int_{-\infty}^{+\infty} dp |\psi_p^F\rangle \phi_{in}(p) - 2\pi i \int_{-\infty}^{+\infty} dp |\psi_p^F\rangle \int_{-\infty}^{+\infty} dp' \frac{m}{|p|} \delta(p+p') \langle \psi_p^F | \hat{V} | \psi_{p'}^F \rangle \phi_{in}(p') \\ &- 2\pi i \int_{-\infty}^{+\infty} dp |\psi_p^F\rangle \int_{-\infty}^{+\infty} dp' \frac{m}{|p|} \delta(p-p') \langle \psi_p^F | \hat{V} | \psi_{p'}^F \rangle \phi_{in}(p') \end{aligned}$$

$$\begin{aligned} \delta\left(\frac{p^2}{2m} - \frac{p'^2}{2m}\right) &= \delta\left(\frac{1}{2m}(p+p')(p-p')\right) = \\ &= \frac{m}{|p|} \delta(p-p') + \frac{m}{|p|} \delta(p+p') \end{aligned}$$

$$\begin{aligned} \int_{-\infty}^{+\infty} d\xi f(\xi) \delta(\xi) &= f(0) \\ \int_{-\infty}^{+\infty} d\xi f(\xi) \delta(\xi - a) &= f(a) \\ \int_{-\infty}^{+\infty} d\xi f(\xi) \delta(\xi - a) &= f(a) \quad \begin{aligned} \xi &= \xi - a \\ d\xi &= d\xi \end{aligned} \\ \int_{-\infty}^{+\infty} d\xi f(\xi) \delta(a\xi) &= a > 0 \quad \begin{aligned} \xi &= a\xi \\ d\xi &= a d\xi \end{aligned} \\ \int_{-\infty}^{+\infty} d\xi \frac{d\xi}{a} f\left(\frac{\xi}{a}\right) \delta(\xi) &= \frac{1}{a} f(0) \\ \delta(a\xi) &= \frac{1}{a} \delta(\xi) \quad a > 0 \\ \delta(a\xi) &= \frac{1}{|a|} \delta(\xi) \quad a \neq 0 \\ \delta(f(\xi)) &= ? \end{aligned}$$



$$\delta(F(\xi)) = \sum_v \frac{1}{|F'(\xi_v)|} \delta(\xi - \xi_v)$$

$$\begin{aligned}
 |\phi_{out}\rangle &= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle \langle \psi_{p'}^F | \hat{S} | \psi_p^F \rangle \langle \psi_{p'}^F | \phi_{in}\rangle = \\
 &= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle \delta(p-p') \phi_{in}(p) \\
 &+ \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle 2\pi i \left(\frac{p^2}{2m} - \frac{p'^2}{2m} \right) \langle \psi_p^F | \hat{V} | \psi_{p'}^+ \rangle \phi_{in}(p') = \\
 &= \int_{-\infty}^{+\infty} dp |\psi_p^F\rangle \phi_{in}(p) \\
 &- \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle \frac{m}{|p|} \delta(p-p') \langle \psi_p^F | \hat{V} | \psi_{p'}^+ \rangle \phi_{in}(p') \\
 &- 2\pi i \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dp' |\psi_p^F\rangle \frac{m}{|p|} \delta(p+p') \langle \psi_p^F | \hat{V} | \psi_{p'}^+ \rangle \phi_{in}(p') = \\
 &\int_{-\infty}^{+\infty} dp = \int_{+\infty}^{-\infty} (-d\tilde{p}) = \tilde{p} = -p \\
 &= \int_{-\infty}^{+\infty} d\tilde{p}
 \end{aligned}$$

$$\begin{aligned}
 &= \int_{-\infty}^{+\infty} dp |\psi_p^F\rangle \phi_{in}(p) \\
 &\left\{ 1 - 2\pi i \frac{m}{|p|} \langle \psi_p^F | \hat{V} | \psi_p^+ \rangle \right\} T(p)
 \end{aligned}$$

$$\begin{aligned}
 &+ \int_{-\infty}^{+\infty} dp |\psi_{-p}^F\rangle \phi_{in}(+p) \\
 &\left\{ -2\pi i \frac{m}{|p|} \langle \psi_p^F | \hat{V} | \psi_p^+ \rangle \right\} = \\
 &R(p)
 \end{aligned}$$

$$\begin{aligned}
 |\phi_{out}\rangle &= e^{-\frac{i}{\hbar} \frac{p^2}{2m} t} \\
 &= \int_{p_{min}}^{p_{max}} dp |\psi_p^F\rangle \phi_{in}(p) T(p) \quad |\phi_{TRANSMITTED}\rangle \\
 &+ \int_{p_{min}}^{p_{max}} dp |\psi_{-p}^F\rangle \phi_{in}(p) R(p) \\
 &e^{-\frac{i}{\hbar} \frac{p^2}{2m} t} \\
 &|\phi_{REFLECTED}\rangle
 \end{aligned}$$

$$|\phi_{out}\rangle = |\phi_{TRANS}\rangle + |\phi_{REFL}\rangle$$

$$|\phi_{TRANS}\rangle = \int_{p_{min}}^{p_{max}} dp |\psi_p^F\rangle \phi_{in}(p) T(p)$$

$$|\phi_{REFL}\rangle = \int_{p_{min}}^{p_{max}} dp |\psi_{-p}^F\rangle \phi_{in}(p) R(p)$$

SQUARE
INTEGRABLE
WAVEPACKETS

$$\langle \phi_{TRANS} | \phi_{REFL} \rangle = 0$$

$$\langle \phi_{TRANS} | \phi_{TRANS} \rangle = \int_{p_{min}}^{p_{max}} dp |\phi_{in}(p) T(p)|^2$$

$$\langle \phi_{REFL} | \phi_{REFL} \rangle = \int_{p_{min}}^{p_{max}} dp |\phi_{in}(p) R(p)|^2$$

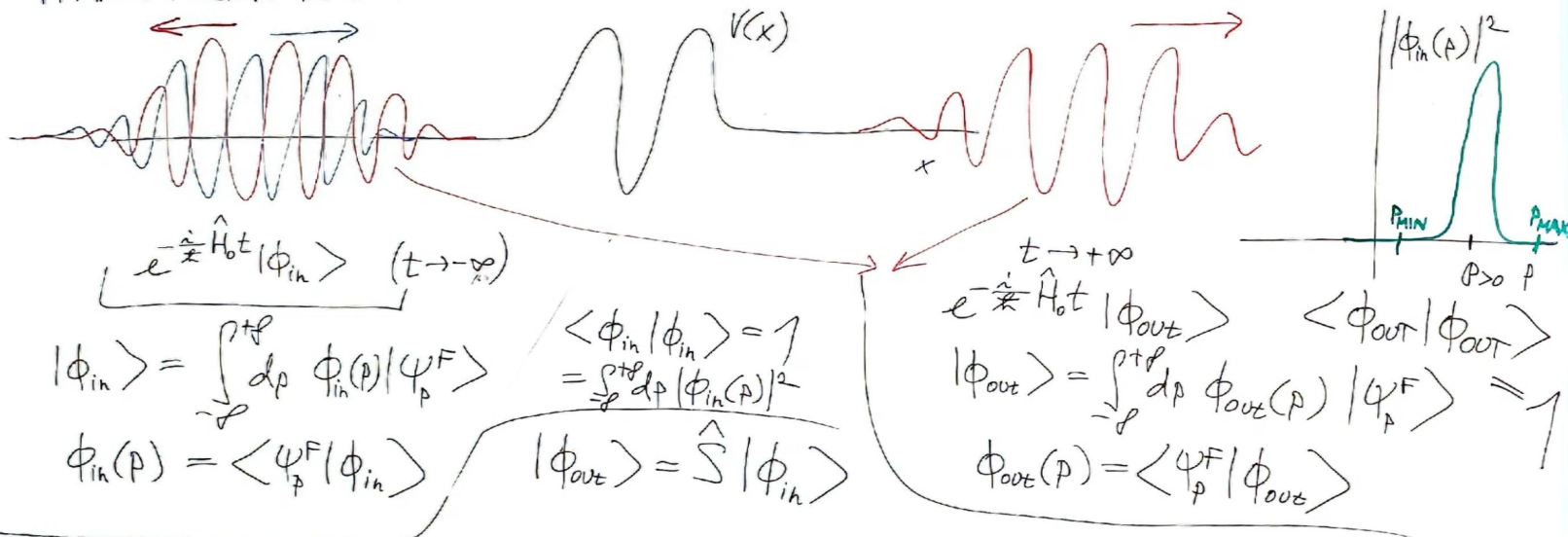
$$\begin{aligned}
 \langle \phi_{out} | \phi_{out} \rangle &= \langle \phi_{TRANS} | \phi_{TRANS} \rangle \\
 &+ \langle \phi_{REFL} | \phi_{REFL} \rangle = 1
 \end{aligned}$$

$$\begin{aligned}
 &= \int_{-\infty}^{+\infty} dp |\psi_p^F\rangle \phi_{in}(p) \\
 &\left\{ 1 - 2\pi i \frac{m}{|p|} \langle \psi_p^F | \hat{V} | \psi_p^+ \rangle \right\} T(p)
 \end{aligned}$$

$$\begin{aligned}
 &+ \int_{-\infty}^{+\infty} dp |\psi_{-p}^F\rangle \phi_{in}(+p) \\
 &\left\{ -2\pi i \frac{m}{|p|} \langle \psi_p^F | \hat{V} | \psi_p^+ \rangle \right\} = \\
 &R(p)
 \end{aligned}$$

$$\begin{aligned}
 |\phi_{out}\rangle &= e^{-\frac{i}{\hbar} \frac{p^2}{2m} t} \\
 &= \int_{p_{min}}^{p_{max}} dp |\psi_p^F\rangle \phi_{in}(p) T(p) \quad |\phi_{TRANSMITTED}\rangle \\
 &+ \int_{p_{min}}^{p_{max}} dp |\psi_{-p}^F\rangle \phi_{in}(p) R(p) \\
 &e^{-\frac{i}{\hbar} \frac{p^2}{2m} t} \\
 &|\phi_{REFLECTED}\rangle
 \end{aligned}$$

TRANSMISSION AND REFLECTION PHENOMENA



$$\langle \psi_p^F | \hat{S} | \psi_{p'}^F \rangle = \delta(p - p') - 2\pi i \delta\left(\frac{p^2}{2m} - \frac{p'^2}{2m}\right) \langle \psi_p^F | \hat{T}(E) | \psi_{p'}^F \rangle$$

$$\hat{T}(E) = \hat{V} + \hat{V} \frac{1}{E - \hat{H} + i\epsilon} \hat{V}$$

$$E = \frac{p^2}{2m} = \frac{p'^2}{2m}$$

$$\langle \psi_p^F | \hat{V} | \psi_{p'}^F \rangle$$

$$|\phi_{out}\rangle = |\phi_{trans}\rangle + |\phi_{refl}\rangle$$

$$|\phi_{trans}\rangle = \int_{p_{MIN}}^{p_{MAX}} dp |\psi_p^F\rangle \phi_{in}(p) T(p)$$

$$|\phi_{refl}\rangle = \int_{p_{MIN}}^{p_{MAX}} dp |\psi_{-p}^F\rangle \phi_{in}(p) R(p)$$

$$\langle \phi_{trans} | \phi_{refl} \rangle = 0$$

$$\langle \phi_{trans} | \phi_{trans} \rangle = \int_{p_{MIN}}^{p_{MAX}} dp |\phi_{in}(p) T(p)|^2$$

$$\langle \phi_{refl} | \phi_{refl} \rangle = \int_{p_{MIN}}^{p_{MAX}} dp |\phi_{in}(p) R(p)|^2$$

$$\langle \phi_{out} | \phi_{out} \rangle = \langle \phi_{trans} | \phi_{trans} \rangle + \langle \phi_{refl} | \phi_{refl} \rangle = 1$$

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WAVEPACKETS

$$= \int_{-\infty}^{+\infty} dp |\psi_p^F\rangle \phi_{in}(p) \left\{ 1 - 2\pi i \frac{m}{|p|} \langle \psi_p^F | \hat{V} | \psi_p^F \rangle \right\} T(p)$$

$$+ \int_{-\infty}^{+\infty} dp |\psi_{-p}^F\rangle \phi_{in}(p) \left\{ -2\pi i \frac{m}{|p|} \langle \psi_p^F | \hat{V} | \psi_p^F \rangle \right\} R(p)$$

$$|\psi_p^F\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} = R(p)$$

$$|\phi_{OUT}\rangle = |\phi_{TRANS}\rangle + |\phi_{REFL}\rangle$$

$$|\phi_{TRANS}\rangle = \int_{p_{MIN}}^{p_{MAX}} dp |\psi_p^F\rangle \phi_{in}(p) T(p)$$

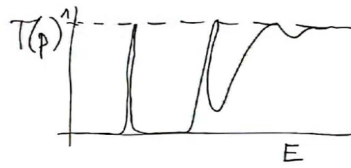
$$|\phi_{REFL}\rangle = \int_{p_{MIN}}^{p_{MAX}} dp |\psi_{-p}^F\rangle \phi_{in}(p) R(p)$$

$$\langle \phi_{TRANS} | \phi_{REFL} \rangle = 0$$

$$\langle \phi_{TRANS} | \phi_{TRANS} \rangle = \int_{p_{MIN}}^{p_{MAX}} dp |\phi_{in}(p) T(p)|^2$$

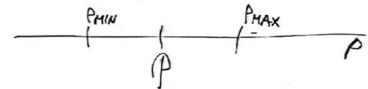
$$\langle \phi_{REFL} | \phi_{REFL} \rangle = \int_{p_{MIN}}^{p_{MAX}} dp |\phi_{in}(p) R(p)|^2$$

$$\langle \phi_{OUT} | \phi_{OUT} \rangle = \langle \phi_{TRANS} | \phi_{TRANS} \rangle + \langle \phi_{REFL} | \phi_{REFL} \rangle = 1$$



SQUARE
INTEGRATE
WAVEPACKETS

CHOOSING NARROW ENOUGH INTERVAL



$T(p)$ AND $R(p)$ ARE ALMOST CONSTANT

$$|\phi_{TRANS}\rangle = T(p) \int_{p_{MIN}}^{p_{MAX}} dp |\psi_p^F\rangle \phi_{in}(p)$$

$$|\phi_{REFL}\rangle = R(p) \int_{p_{MIN}}^{p_{MAX}} dp |\psi_{-p}^F\rangle \phi_{in}(p)$$

$$|T(p)|^2 \int_{p_{MIN}}^{p_{MAX}} dp |\phi_{in}(p)|^2$$

$$|R(p)|^2 \quad |T(p)|^2 + |R(p)|^2 = 1$$

$|\phi_{OUT}\rangle = |\phi_{TRANS}\rangle + |\phi_{REFL}\rangle$

$|\phi_{TRANS}\rangle = \int_{p_{min}}^{p_{max}} |\psi_p^F\rangle \phi_{in}(p) T(p)$

$|\phi_{REFL}\rangle = \int_{p_{min}}^{p_{max}} |\psi_p^F\rangle \phi_{in}(p) R(p)$

SQUARE INTEGRABLE WAVEPACKETS

$\langle \phi_{TRANS} | \phi_{REFL} \rangle = 0$

$\langle \phi_{TRANS} | \phi_{TRANS} \rangle = \int_{p_{min}}^{p_{max}} |\phi_{in}(p) T(p)|^2$

$\langle \phi_{REFL} | \phi_{REFL} \rangle = \int_{p_{min}}^{p_{max}} |\phi_{in}(p) R(p)|^2$

$\langle \phi_{OUT} | \phi_{OUT} \rangle = \langle \phi_{TRANS} | \phi_{TRANS} \rangle + \langle \phi_{REFL} | \phi_{REFL} \rangle = 1$

CHOOSING NARROW ENOUGH INTERVAL

$T(p)$ and $R(p)$ ARE ALMOST CONSTANT

$|\phi_{TRANS}\rangle = T(p) \int_{p_{min}}^{p_{max}} |\psi_p^F\rangle \phi_{in}(p)$

$|\phi_{REFL}\rangle = R(p) \int_{p_{min}}^{p_{max}} |\psi_p^F\rangle \phi_{in}(p)$

$|T(p)|^2$

$|R(p)|^2$

$|T(p)|^2 + |R(p)|^2 = 1$

